SUBCHAPTER 3.3 · MOMENTUM DYNAMICS

Conservation of Linear Momentum & Reaction Forces

Master 5-term balance breakdown, the atmospheric gauge pressure cancellation trick, fluid vs solid reactions, and the 5-stage exam algorithm.

1. The Master Linear Momentum Equation

Newton's Second Law for a closed material fluid system $V_f(t)$ states that the rate of change of linear momentum equals the resultant vector sum of all external forces (surface forces and body forces) acting on the fluid:

$$\frac{d}{dt}\left[\int_{V_f(t)} \rho\vec{v}\, dV\right] = -\int_{\Sigma_f(t)} p\vec{n}\, d\sigma + \int_{\Sigma_f(t)} \bar{\bar{\tau}}' \cdot \vec{n}\, d\sigma + \int_{V_f(t)} \rho\vec{f}_m\, dV$$

[Eq. 4.2] Newton's second law for a closed material fluid system, with $\bar{\bar{\tau}}$ expanded by Eq. 3.28 (Eq. 3.34 is the same law written with $\bar{\bar{\tau}}\cdot\vec{n}$).

Substituting intensive linear momentum $\vec{b} = \vec{v}$ into the universal Reynolds Transport Theorem (Eq. 3.8) yields the **Master Linear Momentum Conservation Equation** for any moving control volume $V_c(t)$:

$$\mathbf{\frac{d}{dt}\left[\int_{V_c(t)} \rho\vec{v}\, dV\right] + \int_{\Sigma_c(t)} \rho\vec{v}[(\vec{v}-\vec{v}_c)\cdot\vec{n}]\, d\sigma = -\int_{\Sigma_c(t)} p\vec{n}\, d\sigma + \int_{\Sigma_c(t)} \bar{\bar{\tau}}'\cdot\vec{n}\, d\sigma + \int_{V_c(t)} \rho\vec{f}_m\, dV}$$

[Eq. 3.35] Master Integral Linear Momentum Equation for an arbitrary moving control volume.

RIGOROUS TERM-BY-TERM PHYSICAL ANATOMY

1. $\frac{d}{dt}\int_{V_c}\rho\vec{v}dV$**:** Unsteady accumulation rate of linear momentum stored inside the geometric boundaries of V_c. (Identically zero for steady flows).
2. $\int_{\Sigma_c}\rho\vec{v}[(\vec{v}-\vec{v}_c)\cdot\vec{n}]d\sigma$**:** Net convective flux of linear momentum carried out of the control volume across Σ_c by the relative velocity $(\vec{v}-\vec{v}_c)$.
3. $-\int_{\Sigma_c} p\vec{n}d\sigma$**:** Net normal pressure force exerted on the control surface. (The negative sign reflects the fact that pressure compresses inward, opposite to outward normal $\vec{n}$).
4. $\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\vec{n}d\sigma$**:** Net viscous shear force (skin friction and viscous stresses) exerted on Σ_c. Often negligible across open flow apertures at high Reynolds numbers.
5. $\int_{V_c}\rho\vec{f}_m dV$**:** Net volumetric body force acting on fluid elements inside V_c (gravity $\rho\vec{g}$, plus inertial forces if non-inertial reference frame).

Linear Momentum: Complete Derivation Chain (Notes.pdf, Eqs. 3.34-3.36 and 4.9-4.10)

Step 1: Newton's second law for a fluid volume (Eq. 3.34)

The surface force on a fluid volume is the integral of the stress vector $\vec{f}_n = \bar{\bar{\tau}} \cdot \vec{n}$ (Eqs. 3.19 and 3.22) and the volume force is $\rho \vec{f}_m \, dV$ (Eq. 3.12). Newton's second law gives Eq. 3.34 in terms of the full stress tensor ($\bar{\bar{\tau}} \cdot \vec{n}$ form):

$$\frac{d}{dt} \int_{V_f(t)} \rho \vec{v} \, dV = \int_{\Sigma_f(t)} \bar{\bar{\tau}} \cdot \vec{n} \, d\sigma + \int_{V_f(t)} \rho \vec{f}_m \, dV$$

Eq. 3.34

Insert the decomposition $\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$ (Eq. 3.28). Because $(p\bar{\bar{I}}) \cdot \vec{n} = p\vec{n}$,

$$\int_{\Sigma} \bar{\bar{\tau}} \cdot \vec{n} \, d\sigma = -\int_{\Sigma} p \, \vec{n} \, d\sigma + \int_{\Sigma} \bar{\bar{\tau}}' \cdot \vec{n} \, d\sigma$$

so the right-hand side splits into a pressure force, a viscous force and a body force. This is the form of the right-hand side of Eq. 3.35 (and of Eq. 4.2).

Step 2: Transfer to a moving control volume (Eq. 3.35)

Eq. 3.8 holds for each Cartesian component of the vector density $\vec{\phi} = \rho \vec{v}$:

$$\frac{d}{dt} \int_{V_f(t)} \rho \vec{v} \, dV = \frac{d}{dt} \int_{V_c(t)} \rho \vec{v} \, dV + \int_{\Sigma_c(t)} \rho \vec{v} \left[(\vec{v} - \vec{v}_c) \cdot \vec{n} \right] d\sigma$$

At the instant considered $V_c = V_f$ and $\Sigma_c = \Sigma_f$, so the force integrals of Step 1 can be evaluated on Σ_c and V_c. Equating the right-hand side of Step 1 to the right-hand side above and moving the convective flux to the other side gives Eq. 3.35:

$$\frac{d}{dt} \int_{V_c(t)} \rho \vec{v} \, dV + \int_{\Sigma_c(t)} \rho \vec{v} \left[(\vec{v} - \vec{v}_c) \cdot \vec{n} \right] d\sigma = -\int_{\Sigma_c(t)} p \, \vec{n} \, d\sigma + \int_{\Sigma_c(t)} \bar{\bar{\tau}}' \cdot \vec{n} \, d\sigma + \int_{V_c(t)} \rho \vec{f}_m \, dV$$

Eq. 3.35

Step 3: Fixed control volume (Eq. 3.36)

For a fixed volume $\vec{v}_c = \vec{0}$, the domain is constant and d/dt passes inside the integral (see the mass section). The convective term becomes $\rho\vec{v}(\vec{v}\cdot\vec{n})$:

$$\int_{V_0} \frac{\partial(\rho\vec{v})}{\partial t}\, dV + \int_{\Sigma_0} \rho\vec{v}\,(\vec{v}\cdot\vec{n})\, d\sigma = -\int_{\Sigma_0} p\,\vec{n}\, d\sigma + \int_{\Sigma_0} \bar{\bar{\tau}}'\cdot\vec{n}\, d\sigma + \int_{V_0} \rho\vec{f}_m\, dV$$

Eq. 3.36

Step 4: Reduction to uniform ports (extension)

Extension, not in the Notes. On a port A_k of a fixed control volume where ρ_k and $\vec{v}_k$ are uniform and $\vec{v}_k$ is normal to the port, the momentum flux integral is evaluated without approximation:

$$\int_{A_k} \rho\vec{v}\,(\vec{v}\cdot\vec{n})\, d\sigma = \rho_k\vec{v}_k\,(\vec{v}_k\cdot\vec{n})\, A_k = \dot{m}_k\,\vec{v}_k, \qquad \dot{m}_k = \rho_k(\vec{v}_k\cdot\vec{n})A_k$$

where $\dot{m}_k$ is negative at inlets (algebraic convention of the mass section). On impermeable walls $\vec{v}\cdot\vec{n} = 0$ and the flux vanishes. For steady flow Eq. 3.36 becomes

$$\sum_k \dot{m}_k\vec{v}_k = \sum_{\text{out}} |\dot{m}|\,\vec{v} - \sum_{\text{in}} |\dot{m}|\,\vec{v} = -\int_{\Sigma_0} p\,\vec{n}\, d\sigma + \int_{\Sigma_0} \bar{\bar{\tau}}'\cdot\vec{n}\, d\sigma + \int_{V_0} \rho\vec{f}_m\, dV$$

This is the "outgoing minus incoming momentum flux equals the sum of forces" statement used in the five-stage algorithm. If the velocity profile at a port is not uniform, the true flux $\int \rho v^2\, dA$ exceeds $\dot{m}\,\bar{v}$ and a momentum-flux correction coefficient is needed.

Step 5: Differential momentum equation (Eq. 4.9) and consistency with Eq. 3.25

Apply Eq. 3.6 to Eq. 4.2 (Eq. 3.34 with Eq. 3.28 inserted) with $\phi = \rho v_i$, and use Gauss' theorem for every surface integral: $\int_\Sigma p\, n_i\, d\sigma = \int_V \partial p/\partial x_i\, dV$, $\int_\Sigma \rho v_i v_j n_j\, d\sigma = \int_V \partial(\rho v_i v_j)/\partial x_j\, dV$ and $\int_\Sigma \tau'_{ij} n_j\, d\sigma = \int_V \partial\tau'_{ij}/\partial x_j\, dV$. The volume is arbitrary, so the arbitrary-volume argument of the mass section gives

$$\frac{\partial(\rho\vec{v})}{\partial t} + \nabla\cdot(\rho\vec{v}\vec{v}) = -\nabla p + \nabla\cdot\bar{\bar{\tau}}' + \rho\vec{f}_m$$

Eq. 4.9

Expand component i of the left-hand side with the product rule:

$$\frac{\partial(\rho v_i)}{\partial t} + \frac{\partial(\rho v_i v_j)}{\partial x_j} = v_i \left[\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v_j)}{\partial x_j}\right] + \rho \left[\frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j}\right] = 0 + \rho \frac{Dv_i}{Dt}$$

The first bracket vanishes by the continuity equation (Eq. 4.5). Hence $\rho\, D\vec{v}/Dt = -\nabla p + \nabla \cdot \bar{\bar{\tau}}' + \rho \vec{f}_m$ (Eq. 4.10), which is Eq. 3.25 with $\nabla \cdot (-p\bar{\bar{I}}) = -\nabla p$. The integral law (Eq. 3.35) and Cauchy's local law (Eq. 3.25) are therefore the same statement.

2. The Gauge Pressure Trick & Solid vs Fluid Conventions

In almost every practical engineering problem, the fluid system operates under ambient atmospheric pressure p_a. If we evaluate absolute pressure p, we must integrate large constants over complex geometries. The **Gauge Pressure Trick** eliminates this burden entirely.

Mathematical Proof

Consider a closed control surface Σ_c. By Gauss's Divergence Theorem, integrating any uniform constant scalar field p_a over a closed boundary yields:

$$\oint_{\Sigma_c} p_a \vec{n}\, d\sigma = \int_{V_c} \nabla(p_a)\, dV = \vec{0} \qquad (\text{since } \nabla p_a = \vec{0})$$

Because this closed integral is strictly zero, we can subtract it from the absolute pressure integral without altering the net force balance in any way:

$$-\int_{\Sigma_c} p\vec{n}\, d\sigma = -\int_{\Sigma_c} p\vec{n}\, d\sigma + \oint_{\Sigma_c} p_a \vec{n}\, d\sigma = -\int_{\Sigma_c} (p - p_a)\vec{n}\, d\sigma = -\int_{\mathbf{\Sigma_c}} \mathbf{p}_{\text{gauge}} \vec{\mathbf{n}}\, \mathbf{d}\sigma$$

> THE GAME-CHANGING PRACTICAL ADVANTAGE
>
> At any open discharge aperture or free liquid surface exposed to the atmosphere, $p = p_a \implies p_{\text{gauge}} = 0$. Consequently, **the pressure integral across all free atmospheric jets drops to zero automatically**! Only pressurized boundaries ($p > p_a$) contribute to the surface integral.

Forces on Fluid vs Forces on Solid (Newton's 3rd Law)

When our control surface cuts across solid anchoring bolts, pipe flanges, or engine test stand mounts, the momentum equation calculates the reaction force exerted **by the solid mounting structure on the fluid** ($\vec{R}_{\text{solid}\to\text{fluid}}$):

$$\sum \vec{F}_{\text{on fluid}} = \vec{R}_{\text{solid}\to\text{fluid}} - \int_{\Sigma_c} p_{\text{gauge}}\vec{n}\, d\sigma + \int_{V_c} \rho\vec{g}\, dV$$

By Newton's Third Law of Action-Reaction, the holding force exerted **by the fluid on the solid structure / anchor** is the exact opposite:

$$\vec{\mathbf{F}}_{\text{fluid}\to\text{structure}} = -\vec{\mathbf{R}}_{\text{solid}\to\text{fluid}}$$

3. The 5-Stage Algorithm for ANY Momentum Exam Problem

Follow this exact chronological blueprint to solve every linear momentum exam problem cleanly:

1. **Define Control Volume V_c & Coordinate System**
   Place inlet and outlet boundaries perpendicular to the flow streamlines (so $\vec{v} \parallel \vec{n}$). Cut across structural flanges where reaction forces $\vec{R}$ act. Set up a clear Cartesian coordinate frame (x, y, z).

2. **Establish Continuity (Mass Conservation)**
   Relate inlet and outlet velocities, cross-sectional areas, and mass flow rates: $\dot{m} = \rho v_1 A_1 = \rho v_2 A_2$. Solve for unknown velocities before touching forces.

3. **Compute Net Convective Momentum Flux**
   For 1D streams, evaluate the vector sum: $\sum \dot{m}_{\text{out}}\vec{v}_{\text{out}} - \sum \dot{m}_{\text{in}}\vec{v}_{\text{in}}$. Respect signs carefully according to your coordinate axes.

4. **Inventory All External Forces (Using Gauge Pressures)**
   Include gauge pressure forces on cut fluid surfaces ($p_g A\vec{n}_{\text{inward}}$), unknown structural reaction vector $\vec{R} = (R_x, R_y, R_z)$, and fluid weight $\vec{W} = -m_f g\hat{j}$ if non-negligible.

5 **Project onto Axes & Solve for Unknown Reactions**

Equate $\sum F_x = \dot{M}_{\text{out},x} - \dot{M}_{\text{in},x}$ and $\sum F_y = \dot{M}_{\text{out},y} - \dot{M}_{\text{in},y}$. Invert for R_x, R_y. Finally, invert signs for forces acting on the solid body!

CANONICAL WALKTHROUGH 1 · CL3 SETUP

Reducing 90° Pipe Elbow & Mouthpiece

Problem: Water (ρ) flows steadily through a reducing 90° elbow lying in the vertical plane x-y ($\hat{j}$ pointing up, so gravity acts along $-\hat{j}$). Inlet 1 has area A_1, velocity $v_1\hat{i}$, and gauge pressure p_{g1}. Exit 2 has area $A_2 < A_1$, points vertically $+y$ ($v_2\hat{j}$), and discharges into the atmosphere ($p_{g2} = 0$). Fluid mass inside the bend is m_f. Determine the anchoring force $\vec{R}$ required to hold the elbow stationary.

1 & 2 Continuity & Momentum Fluxes

From mass conservation: $\dot{m} = \rho A_1 v_1 = \rho A_2 v_2 \implies v_2 = v_1 \frac{A_1}{A_2}$.

Net convective momentum flux:

- $\vec{v}_{\text{in}} = v_1\hat{i}$, with $\dot{m}_{\text{in}} = \dot{m}$
- $\vec{v}_{\text{out}} = v_2\hat{j}$, with $\dot{m}_{\text{out}} = \dot{m}$
- Convective flux: $\sum \dot{m}_{\text{out}}\vec{v}_{\text{out}} - \sum \dot{m}_{\text{in}}\vec{v}_{\text{in}} = -\dot{m}v_1\hat{i} + \dot{m}v_2\hat{j}$

3 & 4 Forces Inventory & Projection

External forces acting on the fluid volume:

- Inlet gauge pressure: $\vec{F}_{p1} = -p_{g1}A_1(-\hat{i}) = +p_{g1}A_1\hat{i}$ (compresses inward into V_c).
- Exit gauge pressure: $\vec{F}_{p2} = 0$ (atmospheric discharge).
- Anchoring reaction from pipe bolts: $\vec{R} = R_x\hat{i} + R_y\hat{j}$.
- Weight of fluid: $\vec{W} = -m_f g\hat{j}$.

Projecting onto Cartesian axes:

$$x\text{-axis}: \quad p_{g1}A_1 + R_x = -\dot{m}v_1 \implies \mathbf{R_x = -p_{g1}A_1 - \dot{m}v_1}$$

$$y\text{-axis}: \quad R_y - m_f g = \dot{m}v_2 \implies \mathbf{R_y = \dot{m}v_2 + m_f g}$$

The force exerted *by the fluid on the elbow* is $\vec{F}_{\text{fluid}\to\text{elbow}} = -\vec{R} = (p_{g1}A_1 + \dot{m}v_1)\hat{i} - (\dot{m}v_2 + m_f g)\hat{j}$.

CANONICAL WALKTHROUGH 2 · CL5 SETUP

Testbench Rocket / Turboreactor Thrust

Problem: A jet engine is mounted on a horizontal test bench. Air enters at mass flow rate $\dot{m}_a$ with speed v_a at atmospheric pressure p_a. Fuel is burned at rate $\dot{m}_f$. Combustion gases exhaust through nozzle area A_e at speed v_e with exit static pressure $p_e \neq p_a$. Determine the horizontal thrust F_T transmitted to the test stand anchor.

Analysis

Control Volume & Momentum Balance

Select a control volume tightly enclosing the engine outer casing, cutting across the inlet (area A_a), nozzle exit (area A_e), and the structural load cell pin R_x holding the engine against thrust.

Continuity: $\dot{m}_e = \dot{m}_a + \dot{m}_f$.

Linear momentum balance along the thrust axis $+x$:

$$\sum F_x = \dot{m}_e v_e - \dot{m}_a v_a$$

External forces acting on the fluid/engine system (using ambient pressure p_a as reference):

$$\sum F_x = R_x - (p_e - p_a)A_e$$

Equating forces to momentum flux:

$$R_x - (p_e - p_a)A_e = \dot{m}_e v_e - \dot{m}_a v_a \implies R_x = \dot{m}_e v_e - \dot{m}_a v_a + (p_e - p_a)A_e$$

The net forward propulsive thrust delivered to the aircraft mount is:

$$\mathbf{F}_{\text{thrust}} = \dot{\mathbf{m}}_\mathbf{e}\mathbf{v}_\mathbf{e} - \dot{\mathbf{m}}_\mathbf{a}\mathbf{v}_\mathbf{a} + (\mathbf{p}_\mathbf{e} - \mathbf{p}_\mathbf{a})\mathbf{A}_\mathbf{e}$$

Composed of *momentum thrust* $\dot{m}_e v_e - \dot{m}_a v_a$ and *pressure thrust* $(p_e - p_a)A_e$.

CANONICAL WALKTHROUGH 3 · CL1 SETUP

Airfoil Drag Evaluation from Wake Deficit (Momentum Integral)

Problem: Using the downstream wake deficit profile from Subchapter 3.2 ($v(y) = v_\infty[1 - \Delta(1 - |y|/b)]$ for $|y| \le b$), compute the total aerodynamic drag force D per unit span acting on the airfoil.

Derivation

The Wake Momentum-Deficit Drag Integral

Select a large rectangular control volume of length $2L$ and height $2H > 2b$. Far upstream and downstream, static pressure has recovered to free stream p_∞, so net pressure forces vanish ($\oint (p - p_\infty)\vec{n}\,d\sigma = \vec{0}$).

The drag D is the horizontal force exerted by the fluid on the airfoil (along $+x$, in the flow direction); by action and reaction the fluid receives $-D$ (pushes $-x$ on the fluid, opposing the flow):

$$-D = \int_{\Sigma_c} \rho v_x (\vec{v}\cdot\vec{n})\, d\sigma$$

Convective flux splits into inlet ($-x$), outlet ($+x$), and lateral outflow escaping through top/bottom at speed v_∞:

$$\int_{\Sigma_{\text{out}}} \rho v_x^2 dy + \dot{m}_{\text{lateral}} v_\infty - \int_{\Sigma_{\text{in}}} \rho v_\infty^2 dy = -D$$

Continuity gives the lateral leakage as inlet minus outlet mass flow. The outlet is uniform ($v = v_\infty$) outside the wake, so $\dot{m}_{\text{lateral}} = \int_{-H}^{H} \rho v_\infty dy - \left[\int_{-H}^{-b} \rho v_\infty dy + \int_{-b}^{b} \rho v(y) dy + \int_{b}^{H} \rho v_\infty dy\right] = \rho \int_{-b}^{b} \big(v_\infty - v(y)\big) dy$. (An earlier version of this page subtracted only the wake

flux, $\int_{-H}^{H} \rho v_\infty dy - \int_{-b}^{b} \rho v \, dy$, which omits the uniform outlet flow outside the wake; the final drag is unchanged.) Substituting, the uniform outer portions cancel identically, yielding the universal **Wake Deficit Drag Integral**:

$$\mathbf{D = \rho \int_{-b}^{b} v(y)[v_\infty - v(y)]\, dy}$$

Substitute $v(y) = v_\infty[1 - \delta]$ where $\delta(y) = \Delta(1 - y/b)$:

$$D = 2\rho v_\infty^2 \int_0^b [1 - \delta(y)]\delta(y)\, dy = 2\rho v_\infty^2 \left[\int_0^b \delta(y) dy - \int_0^b \delta^2(y) dy\right]$$

Evaluating each integral with variable substitution $\xi = 1 - y/b$ ($dy = -b d\xi$):

$$\int_0^b \delta(y) dy = \Delta b \int_0^1 \xi d\xi = \frac{\Delta b}{2}, \qquad \int_0^b \delta^2(y) dy = \Delta^2 b \int_0^1 \xi^2 d\xi = \frac{\Delta^2 b}{3}$$

Combining the integrals yields the exact analytical drag per unit span:

$$\mathbf{D = 2\rho v_\infty^2 b \left(\frac{\Delta}{2} - \frac{\Delta^2}{3}\right) = \rho v_\infty^2 b \Delta \left(1 - \frac{2}{3}\Delta\right)}$$

Wake Momentum Integral: Every Step of the Drag Formula

This section expands the derivation of the wake-deficit drag integral of the airfoil example (profile $v(y) = v_\infty[1-\delta(y)]$, $\delta(y) = \Delta(1-|y|/b)$ for $|y| \le b$, and $v = v_\infty$ outside). It is an application of Eq. 3.36 and Eq. 3.11 (extension, not in the Notes). Per unit span, steady incompressible flow.

Step 1: Control volume and momentum equation

Take the fixed rectangle $-L \le x \le L$, $-H \le y \le H$ with $H > b$, from which the airfoil is excluded, so the airfoil surface is part of the boundary. There $\vec{v}\cdot\vec{n} = 0$ (no flux through a solid wall) and the fluid receives the force $-D\,\vec{e}_x$, where D is the drag of the fluid on the airfoil. Pressure equals p_∞ on the outer boundary, whose net force vanishes (the gauge-pressure argument of the text), and viscous stresses and body forces on the outer boundary are neglected. The x component of Eq. 3.36 for steady flow gives

$$-D = \int_{\Sigma_0} \rho\, v_x\, (\vec{v}\cdot\vec{n})\, d\sigma$$

Step 2: Mass flux through each side

Inlet $x = -L$: $\vec{n} = -\vec{e}_x$, $v_x = v_\infty$. Outlet $x = L$: $\vec{n} = +\vec{e}_x$, $v_x = v(y)$ and the wake occupies $|y| \le b$. Top and bottom $y = \pm H$: fluid leaves through them with total mass flow $\dot{m}_{\text{lat}}$ and with axial velocity $v_x \approx v_\infty$ (outside the wake the streamlines are only slightly displaced; this is the assumption of the example). Eq. 3.11 in steady state states that the outflow equals the inflow:

$$\underbrace{\rho v_\infty\, 2H}_{\text{in}} = \underbrace{\rho v_\infty (2H-2b) + \rho\int_{-b}^{b} v\, dy}_{\text{outlet}} + \dot{m}_{\text{lat}} \;\Longrightarrow\; \dot{m}_{\text{lat}} = \rho\int_{-b}^{b} \big(v_\infty - v\big)\, dy$$

The outlet mass flow has been split into the uniform part outside the wake, of total width $2H-2b$, and the wake itself. This is the lateral leakage that a rigid rectangular control volume must include.

Step 3: Momentum flux through each side

The three contributions to the right-hand side of the momentum equation are

$$\text{inlet: } -\rho v_\infty^2\, 2H, \qquad \text{outlet: } \rho v_\infty^2 (2H-2b) + \rho\int_{-b}^{b} v^2\, dy, \qquad \text{lateral: } \dot{m}_{\text{lat}}\, v_\infty$$

Adding them and substituting $\dot{m}_{\text{lat}}$ from Step 2:

$$-D = -2H\rho v_\infty^2 + (2H - 2b)\rho v_\infty^2 + \rho \int_{-b}^{b} v^2 \, dy + \rho v_\infty \int_{-b}^{b} (v_\infty - v) \, dy$$

The terms in H cancel. Using $\int_{-b}^{b} v_\infty^2 \, dy = 2b\, v_\infty^2$ to cancel the remaining constant $-2b\rho v_\infty^2$:

$$-D = \rho \int_{-b}^{b} \left(v^2 - v_\infty v \right) dy \implies D = \rho \int_{-b}^{b} v \left(v_\infty - v \right) dy$$

Step 4: Evaluation of the integral with explicit limits

Since $v = v_\infty(1 - \delta)$ and $v_\infty - v = v_\infty \delta$, the integrand is $v_\infty^2 (1 - \delta)\delta$. It is even in y, so

$$D = 2\rho v_\infty^2 \int_0^b \left(\delta - \delta^2 \right) dy, \qquad \delta(y) = \Delta\left(1 - \frac{y}{b}\right)$$

Substitute $\xi = 1 - y/b$, so that $\delta = \Delta\xi$ and $dy = -b\, d\xi$. The limits are $y = 0 \Rightarrow \xi = 1$ and $y = b \Rightarrow \xi = 0$. Applying Barrow's rule after reversing the limits:

$$\int_0^b \delta \, dy = \int_{\xi=1}^{\xi=0} \Delta\xi \, (-b \, d\xi) = \Delta b \int_0^1 \xi \, d\xi = \Delta b \left[\frac{\xi^2}{2} \right]_0^1 = \frac{\Delta b}{2}$$

$$\int_0^b \delta^2 \, dy = \Delta^2 b \int_0^1 \xi^2 \, d\xi = \Delta^2 b \left[\frac{\xi^3}{3} \right]_0^1 = \frac{\Delta^2 b}{3}$$

Therefore

$$D = 2\rho v_\infty^2 \left(\frac{\Delta b}{2} - \frac{\Delta^2 b}{3} \right) = \rho v_\infty^2 \, b \, \Delta \left(1 - \frac{2}{3}\Delta \right)$$

Checks. Dimensions: $[\rho v_\infty^2 b] = \mathrm{kg\,m^{-3}\,m^2 s^{-2}\,m} = \mathrm{N/m}$, a force per unit span, as required. Limit $\Delta \to 0$ gives $D \to 0$ (no wake, no drag). Limit $\Delta \to 1$ gives $D \to \rho v_\infty^2 b/3$, finite and positive.